REGIS UNIVERSITY

COLLEGE OF COMPUTER & INFORMATION SCIENCES

Faculty Syllabus
Summer 8 week 1 2019

NOTE to Students: This syllabus is the SAME as the syllabus posted on the Regis website, but with your instructor's contact information, personal policies and additional information ***added in GREEN below***. This version supersedes all other syllabi for this course.

Course Number: CS440

Course Title: Computer Organization and Architecture

Name:	**Dr. Dwayne Ockel**
Telephone Number:	720.257.9876
Email:	dockel@regis.edu / dockel@worldclass.regis.edu (from within D2L)
Facilitator Expectations:	I make every attempt to meet the following turnaround times. There may be instances where it takes longer than the times stated here. Forum messages: Within 24 hours Email & phone calls: Within 24-48 hours Homeworks & Exams: Within 1 week

Course Description

CS 440 - COMPUTER ORGANIZATION AND ARCHITECTURE (3). Introduces Machine Architecture with coverage of digital logic, machine level data and instruction representation, ALU design, and organization of the processor datapath and control. Examines performance analysis, memory system hierarchy, pipelining, and communication.

Prerequisite Courses

CS390 Principles of Programming Languages
or
CS370 Assembly Language (only for degree plans prior to Fall 2015)

Please note that CS202/CS210/CS310 are pre-requisites to CS390 (and CS208/CS361/CS362 are pre-requisites to CS370), so knowledge of the subjects covered in those courses will also be assumed. You need to have a working knowledge of the following topics within each course:

1. Knowledge of basic CPU components and data representation in binary (*CS208 or CS202*).

2. Understanding of the Fetch-Decode-Execute instruction cycle and how basic Assembly Language commands execute (*CS208 or CS202*).

3. Practice with problem definition, solution construction, top-down design techniques, algorithmic development, documentation, testing, debugging, and maintenance (*CS361/CS362 or CS210/CS310*).

4. Ability to use control structures including: if and switch statements and while, do-while, and for loops (*CS361 or CS210*) and data structures (*CS362 or CS310*).

5. Understanding of modular code design with functions, procedures, and proper parameter passing, using both pass-by-value and pass-by-reference (*CS362 or CS310*).

Course Overview

The computer lies at the heart of computing. All students of computing should acquire some understanding and appreciation of a computer system's functional components, their characteristics, their performance, and their interactions.

It is important to understand Computer Architecture in order to structure a program so that it runs efficiently on a real machine. And when selecting a system to use, it is important to understand the tradeoff among various components, so you can accurately compare competing systems, and understand technical literature on new computer systems.

This course will cover the basic concepts of Computer Architecture that are important for you to understand, including the CPU control and datapath, memory systems including caching and virtual memory, and input/output subsystems.

WARNING to students:

Many students find ***this course*** to be one of the ***more difficult*** of the courses in the Regis Computer Science curriculum. So please be sure you will have adequate time to devote to this course before taking it.

Course Outcomes

Upon completion of this course, students should be able to:

1. Explain digital logic and its use in digital systems.
   a. Describe the progression of computer architecture from vacuum tubes to VLSI.
   b. Detail the basic building blocks and their role in the historical development of computer architecture.
   c. Use mathematical expressions to describe the functions of simple combinational and sequential circuits.

d. Design a simple circuit using the fundamental building blocks.

2. Understand machine level representation of data.
   a. Explain the reasons for using different formats to represent numerical data.
   b. Explain how negative integers are stored in sign-magnitude and twos-complement representation.
   c. Explain how floating point numbers are stored in IEEE formats.
   d. Convert numerical data from one format to another.
   e. Discuss how fixed-length number representations affect accuracy and precision.
   f. Perform simple mathematical and logical operations upon numerical data.
   g. Describe the internal representation of nonnumeric data.

3. Understand assembly level machine organization.
   a. Explain the organization and functional units of the classical von Neumann machine, and explain how instructions are executed on it.
   b. Summarize how instructions are represented, at both the machine and assembly levels, for a specific instruction set design.
   c. Explain how instruction formats vary, such as addresses per instruction and variable length vs. fixed length formats.
   d. Write simple assembly language program segments.
   e. Demonstrate how fundamental high-level programming constructs are implemented at the machine-language level.

4. Describe the functional organization of a processor datapath and performance enhancements.
   a. Explain a single cycle datapath implementation.
   b. Compare alternative implementations of datapaths (e.g. multicycle).
   c. Discuss the concept of control points and the generation of control signals.
   d. Explain basic instruction level parallelism using pipelining and the major hazards that may occur.
   e. Describe the implementation of hazard detection and use of register-forwarding for data hazard reduction
   f. Explain the concept of branch prediction and how it is used.
   g. Characterize the costs and benefits of prefetching.
   h. Explain speculative execution and identify the conditions that justify it.

5. Describe memory system organization and architecture.
   a. Identify the main types of memory technology.
   b. Explain the effect of memory latency on running time and how the memory hierarchy reduces the effective memory latency.
   c. Describe the principles of memory management.
   d. Describe the role of cache and compare and contrast direct, fully-associative and set-associative cache organizations.
   e. Explain the workings of a system with virtual memory management.

6. Describe interfacing and communication.

a. Explain how interrupts are used to implement I/O control and data transfers.
b. Identify various types of buses in a computer system.
c. Describe data access from a magnetic disk drive and flash drive.

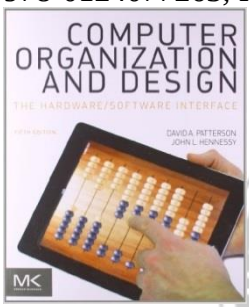
7. Understand technical literature on computer systems.

8. Analyze a computer system's expected performance.
   a. Analyze how each component of a computer system affects performance.
   b. Measure performance using benchmarks and performance metrics.

Course Materials

Required Texts:

Patterson, David A., Hennessy, John L., (2014), *Computer Organization and Design: the Hardware /Software Interface,* (5th Edition). Morgan Kaufmann Publishers, ISBN-13: 978-0124077263, ISBN-10: 0124077269.



NOTE: There is also an "ARM Edition" and "RISC-V Edition" of the Patterson/Hennessy textbook, make sure you get the MIPS version (check the ISBN numbers above)

Technology Tools:

PowerPoint Viewer (online):

Some of the course content is presented via PowerPoint slide shows. If needed, you can download the free PowerPoint viewer via this link:

http://www.microsoft.com/en-us/download/details.aspx?id=13

Pre-Assignment

Read the week 1 reading assignments in your textbook. Be ***prepared*** to ***ask questions*** on unclear areas and ***to respond*** to questions. Sign on to worldclass.regis.edu and become familiar with the course navigation of the Web Curriculum. Complete assignments above.

Course Assignments and Activities

Week	Topics	Readings*	Activities Assignments and Associated Points
1	• Computer Systems and Current Technology • Numbering Systems and Numeric Representations • Arithmetic and Logic Operations and Units	Ch. 1: Sec 1.1 – 1.5 Ch. 2: Sec 2.1 – 2.4 Ch. 3: Sec 3.1 – 3.3, page 196 to top of 211 of Sec 3.5 Appendix B: Sec B.1 – B.3	Participation in Discussions 12% for entire course
2	• Arithmetic and Logic Operations and Units (continued) • Instruction Set Architecture	Ch. 2: Sec 2.5 – 2.7, 2.10 – 2.12, and 2.19 – 2.20 Optional: Sec 2.16 – 2.18	Participation in Discussions Exam #1 16%
3	• Instruction Processing and the Datapath (single cycle) • Performance Analysis	Appendix B Sec B.5 pages B-26 – B-35, Sec B.7 all, and Sec B.8, pages B-50 – B-56 Ch 1, Sec 1.6 Ch 4: Sec 4.1 – 4.4	Participation in Discussions / Project Overview
4	• Instruction Processing and the Datapath (multi-cycle) • Performance Analysis	Appendix B Sec B.5 pages B-26 – B-35, Sec B.7 all, and Sec B.8, pages B-50 – B-56 Ch 1, Sec 1.6 Ch 4: Sec 4.1 – 4.4	Participation in Discussions Exam #2 16%
5	• Pipelining	Ch 4: Sec 4.5 – 4.8, 4.13 – 4.14 Optional: Sec 4.11	Participation in Discussions Project Draft
6	• Memory Hierarchy	Appendix B: Sec B.9 Ch 5: Sec 5.1 – 5.8 and 5.15 – 5.16	Participation in Discussions Exam #3 16%
7	• Input and Output (I/O)	Ch 5: 5.11	Participation in Discussions Final Project 24%
8		None	Participation in Discussions Final Exam 16%
		Total	**100%**

****Previous Edition Reading Assignments:*** Alternate reading assignments for the previous edition of the textbook (4th edition) are available in the Appendix of this syllabus.

Summary of Assignments and Percentage Weight towards course grade

Assignment	Value (percent of overall course grade)
Participation	**12%**
Exams (4 exams at 16% each)	**64%**
Final Project (Paper and Presentation)	**24%**
Totals	100 %

Exams

There will be 4 exams. Exam questions will be cumulative, taken from reading assignments and course content. ***Exams will not be accepted late under any circumstances. A minimum of a four-day window is provided in which to take the exam, so please plan accordingly.***

Final Project

Each student will submit a final project, covering the details of one specific architecture.

Late Assignment Policy for CS440 Final Projects

Late final projects will be graded and then 3**%** will be deducted for each day the assignment is late.

However, final projects will NOT be accepted more than 5 days after the official due date. Therefore, any final projects turned in more than 5 days late will be given a grade of **zero**, and no feedback will be given.

NOTE: The final project consists of both the paper and the presentation. These cannot be submitted separately. If you submit the paper without the presentation, that is your final submission, you cannot submit the presentation separately later.

Topics:

Regarding final project topics… Before submitting your preferences, please do an initial literature review to make sure that sufficient research material exists to complete a research paper of the prescribed length and content. One of the problems with newer or highly proprietary architectures is that there is very little material published beyond typical marketing materials. You need to make sure that you have access to technical content that you can cite in your paper. In general, a **minimum** of around 5-7 reputable sources are needed to produce a quality paper. Once you have received approval for your topic, you are responsible for ensuring that you have sufficient research material. Also, keep in mind that only reputable sources (company whitepapers and data sheets, peer-reviewed journal articles, expert opinion/analysis, etc.) are acceptable. Sources like Wikipedia, most blogs, editorial material, etc. are not acceptable academic sources. Regis' library can be a valuable resource in helping you find and assess the credibility of research material. I highly suggest you engage them early for any help you need regarding help with your research needs, citation assistance, obtaining materials, etc.

Group Projects:

NOTE: Depending on the size of the class, the project may be done in groups instead of individually. If this is the case, special instructions and new requirements will be posted in the course. If this is the case, those requirements replace the requirements within the course content. You still need to submit your preferences as described in the course so that I can make pairings for the groups based on similar interest, where possible.

Participation

Class participation/effort is important because we can all learn from each other. Your participation points can make a difference in the final grade. Participation means:

1. Present in the forum every week
2. Regularly checks forum and post all required assignments/discussion questions/ items by the deadlines.
3. Interacts/replies to other students in forum discussions.
4. Materially and substantively contributing to the discussions with each post. Affirmation postings such as "good job" or "I agree", etc. are not material or substantive and only serve to clutter the forum.
5. Keeping track of all posts and making sure to follow-up on any questions or comments posted by other students.
6. Always posting a responsible manner, practicing appropriate "Netiquette" and being respectful of other participants. Abusive student behavior (either toward other students or toward the instructor) will not be tolerated and appropriate action will be taken as necessary to deal with it.

Please note that forum discussions in this course are used to check your understanding of specific concepts, and to help prepare you for the exams. Therefore, you should think of the discussion questions as a type of "homework", rather than simply "discussions".

Weekly Discussion points distribution:

6 points per Introduction of a Forum Topic (x7),
per Weekly Feedback and Course Feedback response (x8),
and week 3 plagiarism discussion

15 points per correct Discussion Question with Examples response (x9)
per meaningful week 8 Architecture review (x2)
and for the week 3 architecture project overview

24 points for the week 5 final project draft (this is submitted to the Assignments section, not the Discussions section.)

Discussion Question with Example responses will be graded using the rubric below.
Partial points may be given for late, incomplete, or partially incorrect posts.
No points will be given for items posted ***more than one week*** past the due date.

Week 1 there are 42 possible points
Week 2 there are 27 possible points
Week 3 there are 48 possible points
Week 4 there are 27 possible points
Week 5 there are 66 possible points
Week 6 there are 27 possible points
Week 7 there are 27 possible points
Week 8 there are 36 possible points

Total forum points available: **300 pts** = 12% of total course grade

CS440 Discussion Question Rubric						
Demonstration of Subject Matter Comprehension	The initial post ***correctly addressed all parts*** of the discussion question.	There were ***minor*** mistakes in initial post, but the student ***corrected them all*** by the end of the week.	There were ***major*** mistakes in the initial post, but the student ***corrected them all*** by the end of the week.	The initial post contained only ***minor uncorrected*** mistakes, OR there were ***major*** mistakes in the initial post and the student ***only corrected some*** of them by the end of the week.	There were ***major*** mistakes in the initial post, and the student ***did not correct any*** of them by the end of the week.	Student did not post an answer to the discussion question.
Subject Points	15	14	13 - 12	11 - 6	5 - 1	0
Timeliness	Initial post by due date.	Initial answer posted 1 - 7 days late (10% deducted per day late)				No post within 7 days of due date
Late Deductions	0	–10% to –70%				–100% (no credit)

IMPORTANT: Please familiarize yourself with this rubric. There are often questions about discussion grading that are clearly defined in this rubric. For example, the only possible way to receive full credit is if the *initial* post is 100% correct and completely and thoroughly addresses the question/topic at hand. If *any* corrections are needed, you will not receive full credit for the posting. This system is not unique to this course and is being implemented across all courses in the College and again reiterates the concept that the **discussions in an online class are equivalent to homework assignments in a traditional ground-based class**, and are graded accordingly**.** If you turned in a homework assignment that had errors, you wouldn't expect to receive full credit on that homework, the same is true of the discussions in an online course.

Using this rubric and the description of what points are available each week, as described above, along with the late policy, you should be able to ascertain how your discussion grade was arrived at for each week.

NOTE: With regard to the question/comment and weekly feedback posts, points will be awarded at my discretion. The amount of effort put forth will be considered in the grading of these items. Simply because you make a post, does not mean you will receive full credit.

NOTE: Late penalties apply to posts that are initially posted late even if they are originally incorrect but eventually corrected. For example, if you make an initial post a day late, and eventually correct it, you could receive a maximum grade of 12.6 (14 would be the maximum non-penalized grade, with a 10% reduction for being a day late).

Course Policies and Procedures

Adding this course during the Drop/Add Period

If you added this course during the drop/add period, after class began on Monday, you are responsible for ***immediately*** notifying the instructor that you joined the course late. None of the course due dates will be extended for you. Even if a due date already passed when you added the course, late points will still be deducted.

Repeating the course

If you are repeating this course (due to a previous withdraw or low grade), you are responsible for ***immediately*** notifying the instructor. Course assignments that you submitted when you last took the course cannot be repeated -- you will be required to complete alternate assignments.

If you fail to do this and end up submitting duplicate work from the last time you took the course, you will receive a zero for all of that work without the opportunity to resubmit it.

Academic Integrity

Plagiarism includes submitting work obtained from any person, publication, or internet web source. All work submitted in CS440 must be your own.

It is plagiarism to submit work under your own name(s) in which you:

- collaborated with other people in solving the problem
- copied the solution from a book, web site, or the work of another
- got significant help, but do not acknowledge this in your solution

In this course, you are ***not allowed*** to submit any joint work. All submitted work must be your own. You must always provide proper attribution of authorship by naming all persons, books, or resources that provided intellectual content towards the final result. To fail to provide proper attribution is plagiarism. Plagiarism demeans the seriousness of what we do in class, and does not allow you as a student to obtain a fair grade for the results you worked hard for.

In cases of suspected cheating or plagiarism, the instructor will discuss the matter with the student(s) involved. The instructor reserves the right to question any student orally or in writing about any assignment, and to use the evaluation of the student's understanding of the assignment and of the submitted solution as evidence of cheating.

All cheating and plagiarism incidents will be reported to the Academic Integrity Board.

Academic Integrity Board

ALL instances of academic dishonesty will be taken seriously and will be reported to the Academic Integrity Board.

The Board has the authority to issue sanctions/penalties up to and including expulsion from the University. Academic dishonesty includes cheating of any kind, plagiarism, fabrication, collusion, failure to provide proper attribution, and various other unacceptable academic conduct. Please visit https://www.regis.edu/About-Regis-University/Policies-and-Procedures/Academic-Integrity-Policies.aspx to familiarize yourself with policies concerning academic integrity.

Please visit my writing information and resources page at http://academic.regis.edu/dockel/WRITING/writing_main.asp where you will find resources regarding how to avoid plagiarism, paraphrase, and cite sources correctly. Additionally, it contains information/links to Regis resources including writing consultants and library assistance. The final project is a major research paper and any form of plagiarism will result in a punitive response. Penalties for plagiarism can range from receiving a zero on the assignment to expulsion from the university, depending on various factors. Please consult the link above for more information on the penalties and consequences of plagiarism as well as examples of behaviors that constitute academic dishonesty (including cheating, plagiarism, fabrication, collusion, etc.) If you have questions regarding your own writing, please contact a librarian or writing consultant for assistance (see the above URL for links to these resources).

CC&IS Grading Scale

Letter Grade	Percentage	Grade Point
A	93 to 100	4.00
A–	90 to less than 93	3.67
B+	87 to less than 90	3.33
B	83 to less than 87	3.00
B–	80 to less than 83	2.67
C+	77 to less than 80	2.33
C	73 to less than 77	2.00
C–	70 to less than 73	1.67
D+	67 to less than 70	1.33
D	63 to less than 67	1.00
D-	60 to less than 63	.67
F	Less than 60	0

Additional information about grading can be found in the latest edition of the University Catalog, available at http://www.regis.edu/Academics/Course%20Catalog.aspx.

CC&IS Policies and Procedures

Each of the following CC&IS Policies & Procedures is incorporated here by reference. Students are expected to review this information each term, and agree to the policies and procedures as identified here and specified in the latest edition of the University Catalog, available at http://www.regis.edu/Academics/Course%20Catalog.aspx or at the link provided.

- The CC&IS Academic Integrity Policy. The policy is available at: http://www.regis.edu/About-Regis-University/Policies-and-Procedures/Academic-Integrity-Policies.aspx
- The Student Honor Code and Student Standards of Conduct.
- Incomplete Grade Policy, Pass / No Pass Grades, Grade Reports.
- The Information Privacy policy and FERPA. For more information regarding FERPA, visit the U.S. Department of Education.
- The HIPPA policies for protected health information. The complete Regis University HIPAA Privacy & Security policy can be found here: http://www.regis.edu/About-Regis-University/University-Offices-and-Services/Auxiliary-Business/HIPAA.aspx.
- The Human Subjects Institutional Review Board (IRB) procedures. More information about the IRB and its processes can be found here: http://regis.edu/Academics/Academic-Grants/Proposals/Regis-Information/IRB.aspx.

The CC&IS Policies & Procedures Syllabus Addendum summarizes additional important policies including, Diversity, Equal Access, Disability Services, and Attendance & Participation that apply to every course offered by the College of Computer & Information Sciences at Regis University.

A copy of the CC&IS Policies & Procedures Syllabus Addendum can be found here: https://in2.regis.edu/sites/ccis/policies/Repository/CCIS%20Syllabus%20Addendum.docx. (RegisNet username and password required for access)

ADDITIONAL POLICIES:

A Note about Online Exams

Exams 1-3 are automatically graded. However, the automatic grader does not always correctly grade short answer questions due to spacing, punctuation, etc. If you believe a question has been incorrectly scored, please let me know and I will adjust your score as warranted. Additionally, I reserve the right to adjust any scoring on any exam as appropriate, even if it results in a lower overall score. For example, I reserve the right to correct any mistakes the auto-grader may make in your favor as well as those that you got correct but the auto-grader scored as incorrect.

Also, remember that the exams run during a window of time and just because you have finished an exam, doesn't mean everyone else has. Do NOT post any questions/information about the

content of the exams until after the exam window has expired. Doing so can give an unfair advantage to those who have not yet taken the exam and is considered a form of academic dishonesty. You can email me privately if you wish regarding the exams. Extensions will not be given on the exams. The purpose of the window is to allow for any unexpected issues, etc. that might come up. It is highly recommended that you take the exam as early in the window as possible.

Finally, be sure to proofread your exam before submitting it. The auto-grader and instructor can only grade what is submitted. Credit cannot be given for typos. Keep in mind, in the real-world, typos can result in catastrophic consequences and is a leader source of bugs in code. The four-hour window given for the exams is intended to provide enough time for you to go over your exam before submitting it.

Further, I will not argue or negotiate about partial credit that may or may not be applied to specific questions. The exams are setup to award partial credit for specific answers if that is deemed appropriate. In many cases, there is no partial credit for a question. This also goes for alternative exams if you have taken the course before. The alternative exams are setup to mirror the online exams in terms of content, point distribution/credit, etc.

A Note about Rubrics

It is impossible for the grading rubric for the final project to contain/describe every possible scenario in which points could be deducted. The rubric is intended to serve as a **general guideline/template** for grouping deducted points into appropriate categories. However, there may be deductions against a category that are not specifically described in the rubric itself. When points are deducted, I try to subtract points from the most appropriate category. These deductions will be outlined in the comments section of the rubric. Just keep in mind that the rubric is not "all inclusive" in terms of identifying every possible scenario for which points may be deducted. Points may be deducted from a category despite there being no specific mention of the reason for that deduction in the rubric, based on the judgment and discretion of the instructor. Also note that there is always a subjective element in grading certain items such as research papers. I will use my expertise in the subject domain and my professional judgment – there isn't always a black and white answer to a given question, subject, discussion, etc.

Turnitin.com / Feedback Studio

I reserve the right to use the plagiarism detection software at turnitin.com for any papers submitted for this course. Assignments submitted to the submission folder are automatically submitted to Turnitin for plagiarism detection and writing originality scoring. For more information on Turnitin, there are a number of whitepapers available at http://turnitin.com/en_us/resources-old/white-papers as well as an FAQ at https://www.turnitin.com/help_pages/student_faq.asp?r=87.6295589277586

I also recommend http://www.plagiarism.org/ which contains examples, resources, citing information, etc. regarding plagiarism.

Incompletes

I very rarely issue incomplete grades and typically do so only when documented evidence of an extraordinary circumstance is presented (such as a serious illness or death in the family). Additionally, an incomplete will only be considered if you are currently passing the course at the time of the request and have completed at least half of the course work (based on available points from all applicable forum participation, quizzes/exams, papers, homework assignments, etc.) Furthermore, it may not be possible to achieve full credit on the work that must be made up. Incompletes will not be given due to time management issues, employment assignments/obligations, falling behind, etc.

Errors/Misprints, etc.

I reserve the right to make any necessary corrections throughout the course as the result of unintentional errors, misprints, broken links, etc.

Extra Credit

There is no extra credit available for this course.

Student Responsibility

All students are expected to read and fully comply with this syllabus. This syllabus takes precedence over the generic syllabus posted in the course and on the Regis syllabus page. Those syllabi are intended to serve only as templates and general guidelines. Students are also expected to read **all** postings in the forum, especially any in the Faculty Notes forums. It is assumed that students have read all postings in the forums and are familiar with any changes, corrections, or other information that may affect any aspect of the course. Please check in on the course frequently and read all forum postings between asking questions, you may find that your question has already been answered.

APPENDIX

Alternate Readings for 4th Edition of Textbook

Week	Learning Topics	Readings in 4th Edition of Textbook
1	LT 1: Computer Systems and Current Technology LT 2: Numbering Systems and Numeric Representations LT 3: Arithmetic and Logic Operations and Units	Ch. 1: Sec 1.1 – 1.3 and 1.7, pages 44 – 47 only Ch. 2: Sec 2.1 – 2.4 Ch. 3: Sec 3.1 – 3.4 and up to page 258 of Sec 3.5 Appendix C: sec C.1-C.3*
2	LT 3: (continued) LT 4: Instruction Set Architecture	Ch. 2: Sec 2.1 – 2.7, 2.10, 2.12, and 2.19
3	LT 5: Instruction Processing and the Datapath (single cycle) LT 6: Performance Analysis	Appendix C: sec C.5 - C.8 Ch 1, Sec 1.4, bottom of page 26 to middle of page 33 only Ch 4: Sec 4.1 – 4.4
4	LT 5: Instruction Processing and the Datapath (multi-cycle) LT 6: Performance Analysis	Ch 1: Sec 1.4, pages 33-38 Ch 1: Sec 1.5 – 1.9
5	LT 7: Pipelining	Ch 4: Sec 4.5 – 4.8, 4.13 – 4.14
6	LT 8: Memory Hierarchy	Ch 5: Sec 5.1 – 5.5 and 5.10 – 5.12 Appendix C: sec C.9 only
7	LT 9: Input and Output (I/O)	Ch. 6 (all)
8		none

*Download Appendix C via Resources link at:

http://www.elsevierdirect.com/v2/companion.jsp?ISBN=9780123747501